

Tehran Road Network and University Accessibility Analysis

Advanced Programming, HW3
Department of Math & Computer Science, KNTU

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Overview

This project consists of two main parts and is designed to introduce students to practical spatial network analysis using real-world geographic data. The goal is to work with Tehran's road network to first implement an interactive route-finding system and then perform a large-scale accessibility analysis of universities relative to buildings in the city.

Part 1: Interactive Route Finder

In the first part, students will develop a simple application that allows a user to interactively select two points on a map and compute the shortest driving route between them based on real road network data.

Objectives

- Load and preprocess Tehran's road network using the `pyrosm` library.
- Display an interactive map interface to the user (e.g., using `folium`) where the user can click to select a starting and ending location.
- Convert the selected coordinates to the nearest road nodes.
- Compute the shortest route between the selected nodes using real road distances.
- Display the route on the map.

Part 2: University Influence Area Analysis

The second part of the project involves analyzing the “influence area” or accessibility zone of each university in Tehran. The goal is to determine which university is the most accessible to each building in the city.

Objectives

- Extract building and university data from the same OSM file used in Part 1.
- For each building, identify the closest university using network distances (not Euclidean distances) to represent actual road accessibility.
- Use appropriate spatial and network analysis tools (e.g., igraph) to calculate these distances efficiently.
- Assign each building to its closest university and create a thematic map showing the accessibility zones.

Data Requirements

- Tehran OpenStreetMap data in PBF format.

Expected Deliverables

1. A Jupyter Notebook containing both parts of the analysis.
2. Screenshots or exports of the map visualizations.
3. A short report explaining the methodology and findings.